

Time Series Basics

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Time Series Basics

Two different kinds of data:

1. Cross section $(y_i)_{i=1}^n$: Independent & identically distributed
E.g., cross sectional regression

$$\begin{aligned}y_i &= \mathbb{E}[y_i | X_i = x] + u_i \\ &= m(x) + u_i\end{aligned}$$

2. Time series $(y_t)_{t=1}^T$: Temporal ordering, dependencies among $(y_t)_{t=1}^T$
E.g., predictive regression

$$\begin{aligned}y_{t+h} &= \mathbb{E}[y_{t+h} | X_t = x] + u_{t+h} \\ &= m(x) + u_{t+h}\end{aligned}$$

Is $m(x)$ well-defined? We want $m(x)$ to not depend on t . Under what assumptions can we do this?

Strict stationarity

1. Stationarity means invariance with respect to time shift.
2. A process:

$$y_1, y_2, y_3, \dots$$

is strictly stationary if for any $(t_1, \dots, t_p)$, and for any integer h , the joint distribution of $(y_{t_1}, \dots, y_{t_p})$ is equal to the joint distribution of $(y_{t_1+h}, \dots, y_{t_p+h})$.

3. Strict stationarity is a rather stronger concept than identical distribution since it requires all marginal distributions to be identical.

Weak stationarity

Joint finite distributions are difficult to work with and hard to obtain in general. Therefore, we introduce a simple concept of stationarity which only characterizes first two moments of the process.

Definition:

Let $\{y_t\}_{t=-\infty}^{\infty}$ be such that $\mathbb{E}y_t^2 < \infty$ for all t . The autocovariance function of y_t is given by

$$\gamma(t, s) = \mathbb{E}[(y_t - \mathbb{E}y_t)(y_s - \mathbb{E}y_s)]$$

- ▶ $\gamma(t, t) \geq 0$
- ▶ $|\gamma(t, s)| \leq \sqrt{\gamma(t, t)}\sqrt{\gamma(s, s)}$

Weak stationarity

Definition:

The time series $\{y_t\}_{t=-\infty}^{\infty}$ is weakly stationary if

- ▶ $\mathbb{E}y_t^2 < \infty$ for all t
- ▶ $\mathbb{E}y_t = c$ for all t
- ▶ $\gamma(t, s) = \gamma(t + h, s + h)$ for all t, s, h (invariant w.r.t. time shift)

The third property of weakly stationary process implies that

$$\gamma(t, s) = \gamma(t - s, 0)$$

so that the autocovariance function is well defined as only a function of the lag $h = t - s$.

$$\gamma(h) = \gamma(h, 0) = \gamma(h + t, t) = \mathbb{E}[(y_{t+h} - \mathbb{E}y_{t+h})(y_t - \mathbb{E}y_t)]$$

Strict stationarity implies weak stationarity as long as second moments are finite.

Examples

White Noise:

$\{\varepsilon_t\}$ is said to be White Noise if it is serially uncorrelated and has a common variance with mean zero.

$$\varepsilon_t \sim WN(0, \sigma^2)$$

$$\mathbb{E}\varepsilon_t = 0, \forall t = 1, 2, \dots$$

$$\gamma_\varepsilon(h) = \begin{cases} \sigma^2 & h = 0 \\ 0 & h \neq 0 \end{cases}$$

Examples

AR(1) Process:

$$y_t = \phi y_{t-1} + \varepsilon_t, \varepsilon_t \sim WN(0, \sigma^2)$$

- ▶ $|\phi| < 1$: (Covariance) stationary
- ▶ $|\phi| \geq 1$: Nonstationary
- ▶ $\phi = 1$: Unit root
- ▶ $\phi > 1$: Explosive process

How to find the mean and autocovariance of $\{y_t\}$?

Examples

1. Repeated substitution (when ε_t is assumed to exist for all $t = 0, -1, -2, \dots$):

$$y_t = \varepsilon_t + \phi\varepsilon_{t-1} + \phi^2\varepsilon_{t-2} + \phi^3\varepsilon_{t-3} + \dots$$

We have

$$\mathbb{E}y_t = 0$$

and

$$\begin{aligned}\gamma_y(h) &= \mathbb{E}[y_{t+h}y_t] \\ &= \mathbb{E}[(\varepsilon_{t+h} + \phi\varepsilon_{t+h-1} + \phi^2\varepsilon_{t+h-2} + \dots)(\varepsilon_t + \phi\varepsilon_{t-1} + \phi^2\varepsilon_{t-2} \dots)] \\ &= \phi^h\sigma^2 + \phi^{h+2}\sigma^2 + \dots \\ &= \frac{\phi^h\sigma^2}{1 - \phi^2}\end{aligned}$$

if $|\phi| < 1$.

Examples

2. Repeated substitution:

$$y_t = \varepsilon_t + \phi\varepsilon_{t-1} + \phi^2\varepsilon_{t-2} + \dots + \phi^{t-1}\varepsilon_1 + \phi^ty_0$$

We have

$$\mathbb{E}y_t = 0$$

as long as $\mathbb{E}y_0 = 0$. Also,

$$\gamma_y(h) = \mathbb{E}[(y_{t+h} - \mathbb{E}y_{t+h})(y_t - \mathbb{E}y_t)]$$

when is the autocovariance function independent of t ? $\implies y_0$ is uncorrelated with all $\{\varepsilon_s\}_{s=1}^t$ and has variance $\frac{\sigma^2}{1-\phi^2}$.

Examples

3. Assuming stationarity, we have

$$\begin{aligned}y_t &= \phi y_{t-1} + \varepsilon_t \\ \implies \mathbb{E}y_t &= \phi \mathbb{E}y_{t-1} + 0\end{aligned}$$

so that

$$\mathbb{E}y_t = 0$$

Also, multiplying both sides by y_{t-h} and taking expectations,

$$\begin{aligned}\mathbb{E}y_{t-h}y_t &= \phi \mathbb{E}y_{t-h}y_{t-1} + \mathbb{E}y_{t-h}\varepsilon_t \\ \implies \gamma_y(h) &= \phi \gamma_y(h-1) + 0\end{aligned}$$

Squaring both sides and taking expectations, we have

$$\begin{aligned}\mathbb{E}y_t y_t &= \phi^2 \mathbb{E}y_{t-1} y_{t-1} + 0 + \mathbb{E}\varepsilon_t^2 \\ \implies \gamma_y(0) &= \frac{\sigma^2}{1 - \phi^2}\end{aligned}$$

Examples

Estimation is done by either least squares or ML. We have

$$p(y_1, \dots, y_T | \theta) = p(y_1 | \theta) p(y_2 | y_1, \theta) p(y_3 | y_1, y_2, \theta) \dots p(y_t | y_1, y_2, \dots, y_{t-1}, \theta)$$

If we have AR(1) model for y_t , the likelihood can be written as

$$p(y_1, \dots, y_T | \theta) = p(y_1 | \theta) p(y_2 | y_1, \theta) p(y_3 | y_2, \theta) \dots p(y_t | y_{t-1}, \theta)$$

Assuming Gaussian errors $\{\varepsilon_t\}$, we have

$$\log \prod_{t=2}^T p(y_t | y_{t-1}, \theta) = -\frac{T-1}{2} \log(2\pi) - \frac{T-1}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \sum_{t=2}^T (y_t - \phi y_{t-1})^2$$

so that ignoring the initial condition y_1 , maximum likelihood estimator is equal to the least squares estimator.

Examples

Multi-dimensional and multi-lag extension of AR(1) is given by the VAR(p) model

$$y_t = \sum_{i=1}^p A_i y_{t-1} + \varepsilon_t$$

where $A_i, i = 1, \dots, p$ are $n \times n$ dimensional matrix, and $y_t, \varepsilon_t \in \mathbb{R}^n$. ε_t are given by a multivariate White Noise process

$$\varepsilon_t \sim WN(0, \Sigma)$$

$\hat{A}_i, i = 1, \dots, p$ may be obtained using multivariate regression, i.e., equation by equation linear regression. $\hat{\Sigma}$ may be obtained from

$$\hat{\Sigma} = \frac{1}{T-p} \sum_{t=p+1}^T \left(y_t - \sum_{i=1}^p \hat{A}_i y_{t-1} \right) \left(y_t - \sum_{i=1}^p \hat{A}_i y_{t-1} \right)'$$

Examples

MA(1) Process:

$$y_t = \varepsilon_t + \theta\varepsilon_{t-1}, \varepsilon_t \sim WN(0, \sigma^2)$$

MA processes are covariance stationary regardless of the values of θ .

$$\begin{aligned}\mathbb{E}y_t &= 0 \\ \gamma_y(h) &= \mathbb{E}[(\varepsilon_{t+h} + \theta\varepsilon_{t+h-1})(\varepsilon_t + \theta\varepsilon_{t-1})] \\ &= \begin{cases} \sigma^2 + \theta^2\sigma^2 & h = 0 \\ \theta\sigma^2 & |h| = 1 \\ 0 & |h| > 1 \end{cases}\end{aligned}$$

Examples

Assuming $\varepsilon_0 = 0$, we may invert the MA(1) process

$$y_1 = \varepsilon_1 + 0 \implies \varepsilon_1 = y_1$$

$$y_2 = \varepsilon_2 + \theta\varepsilon_1 \implies \varepsilon_2 = y_2 - \theta y_1$$

$$y_3 = \varepsilon_3 + \theta\varepsilon_2 \implies \varepsilon_3 = y_3 - \theta y_2 + \theta^2 y_1$$

so that we have

$$\varepsilon_t = y_t + \sum_{j=1}^{t-1} (-\theta)^j y_{t-j}$$

If $|\theta| < 1$, the MA(1) process is called 'invertible'.

Examples

We may estimate MA(1) using ML. Notice that

$$\log f(y_1, \dots, y_T) = \log f(\varepsilon_1, \dots, \varepsilon_T) = -\frac{T}{2} \log(2\pi) - \frac{T}{2} \log(\sigma^2) - \sum_{t=1}^T \frac{\varepsilon_t^2}{2\sigma^2}$$

where ε_t are given by

$$\varepsilon_t = y_t + \sum_{j=1}^{t-1} (-\theta)^j y_{t-j}$$

This is nonlinear optimization problem that does not have a closed form solution.

What does stationarity give us?

Suppose we have a univariate time series $y_t, t = 1, \dots, T$ that we model as a joint normal random vector. Let object of interest is the random vector $Y = (y_1, \dots, y_T)'$ with mean $\mu = (\mu_1, \dots, \mu_T)'$ and variance

$$\Sigma = \begin{pmatrix} \sigma_1^2 & \sigma_{12} & \dots & \sigma_{1T} \\ \sigma_{21} & \sigma_2^2 & \dots & \sigma_{2T} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{T1} & \sigma_{T2} & \dots & \sigma_T^2 \end{pmatrix}$$

The joint pdf of Y is given by

$$\log p(Y|\theta) = -\frac{T}{2} \log(2\pi) - \frac{1}{2} \log |\Sigma| - \frac{1}{2} (Y - \mu) \Sigma^{-1} (Y - \mu)$$

We only have one sample realization but around $T^2/2$ number of parameters to estimate.

What does stationarity give us?

Stationarity gives us 'parameter sharing'. Because joint distribution of y_1, y_2 is same as that of y_{1+h}, y_{2+h} for all h , we have that

$$\sigma_{21} = \sigma_{32} = \dots = \sigma_{T,T-1}$$

Similarly, we may show that if y_t is stationary process, we must have $\mu_t = \bar{\mu}$, $\sigma_t^2 = \bar{\sigma}^2$ for all t , and that diagonals of Σ should all be the same (Toeplitz matrix).

What does stationarity give us?

Consider the AR(1) example. Here, we have $\mu = (0, \dots, 0)'$ and

$$\Sigma = \begin{pmatrix} \frac{\sigma^2}{1-\phi^2} & \frac{\phi\sigma^2}{1-\phi^2} & \cdots & \frac{\phi^{T-1}\sigma^2}{1-\phi^2} \\ \frac{\phi\sigma^2}{1-\phi^2} & \frac{\sigma^2}{1-\phi^2} & \cdots & \frac{\phi^{T-2}\sigma^2}{1-\phi^2} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\phi^{T-1}\sigma^2}{1-\phi^2} & \frac{\phi^{T-2}\sigma^2}{1-\phi^2} & \cdots & \frac{\sigma^2}{1-\phi^2} \end{pmatrix}$$

so that we are only left with 2 parameters (ϕ, σ^2) to estimate.

What does stationarity give us?

Consider the MA(1) example. Here, we have $\mu = (0, \dots, 0)'$ and

$$\Sigma = \begin{pmatrix} \sigma^2 + \theta^2 \sigma^2 & \theta \sigma^2 & 0 & \dots & 0 \\ \theta \sigma^2 & \sigma^2 + \theta^2 \sigma^2 & \theta \sigma^2 & \dots & 0 \\ 0 & \theta \sigma^2 & \sigma^2 + \theta^2 \sigma^2 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & \dots & \sigma^2 + \theta^2 \sigma^2 & \end{pmatrix}$$

so that we are only left with 2 parameters (θ, σ^2) to estimate.